GraphSAE: Supplementary Material

I. Source-node Contribution Decomposition for Other Backbones

This section extends the GCN decomposition in Section III-C (Source-node Contribution Decomposition) of the main paper to GIN, GAT, and GraphSAGE.

Let $G = (V, E, X)$ be the input graph, $h_v^{(\ell)}$ the representation of node v at layer ℓ, and $h_v = h_v^{(L)}$ its final representation. For a node-level target $\tau = t$, the target representation is $s_\tau = h_t$; for a graph-level target $\tau = G$, it is $s_\tau = q_G = \mathrm{READOUT}(\{h_v : v \in V\})$. Source-node terms are initialized as $r_{u\to v}^{(0)} = \mathbb{I}[u = v]x_u$ and propagated through the frozen backbone. Each nonlinearity uses its observed local linear region. For a ReLU with pre-activation $a_v^{(\ell+1)}$, the gate $g_v^{(\ell+1)} = \mathbb{I}[a_v^{(\ell+1)} > 0]$ multiplies contributions elementwise. Bias contributions are collected in the target-level remainder defined in Section III-C of the main paper.

A. GIN

For a GIN layer, define

$$\xi_v^{(\ell)} = (1+\epsilon^{(\ell)})h_v^{(\ell)} + \sum_{w\in\mathcal{N}(v)} h_w^{(\ell)}, \quad h_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}(\xi_v^{(\ell)}).$$

Let $J_v^{(\ell)}$ be the Jacobian of $\mathrm{MLP}^{(\ell)}$ at $\xi_v^{(\ell)}$. It contains the gates of the MLP's internal nonlinearities and is therefore its local linear map. The immediate neighbor and self messages are $m_{u\to v}^{(\ell+1)} = J_v^{(\ell)} h_u^{(\ell)}$, $m_{v\to v}^{(\ell+1)} = J_v^{(\ell)}(1+\epsilon^{(\ell)})h_v^{(\ell)}$. Applying that same local map to source-specific terms yields

$$r_{u\to v}^{(\ell+1)} = J_v^{(\ell)} \left((1+\epsilon^{(\ell)}) r_{u\to v}^{(\ell)} + \sum_{w\in\mathcal{N}(v)} r_{u\to w}^{(\ell)} \right).$$

B. GAT

For a GAT layer, with attention coefficients retained from the frozen forward pass,

$$h_v^{(\ell+1)} = \sigma\left(\sum_{w\in\widehat{\mathcal{N}}(v)} \alpha_{vw}^{(\ell)} W^{(\ell)} h_w^{(\ell)} \right),$$

where $\widehat{\mathcal{N}}(v)$ includes the self-loop. The immediate contribution from neighbor u is $m_{u\to v}^{(\ell+1)} = \alpha_{vu}^{(\ell)} W^{(\ell)} h_u^{(\ell)}$. Attention coefficients $\alpha_{vw}^{(\ell)}$ are held fixed at their forward-pass values during source decomposition. The source-node recursion is

$$r_{u\to v}^{(\ell+1)} = g_v^{(\ell+1)} \odot \sum_{w\in\widehat{\mathcal{N}}(v)} \alpha_{vw}^{(\ell)} W^{(\ell)} r_{u\to w}^{(\ell)}.$$

C. GraphSAGE

For the mean-aggregator GraphSAGE layers used here, write $W^{(\ell)} = [W_{\mathrm{self}}^{(\ell)}\ W_{\mathrm{nbr}}^{(\ell)}]$. Their update is

$$h_v^{(\ell+1)} = \sigma\left(W_{\mathrm{self}}^{(\ell)} h_v^{(\ell)} + W_{\mathrm{nbr}}^{(\ell)} \frac{1}{|\mathcal{N}(v)|} \sum_{w\in\mathcal{N}(v)} h_w^{(\ell)} \right).$$

The neighbor and self messages are, respectively, $m_{u\to v}^{(\ell+1)} = \frac{1}{|\mathcal{N}(v)|} W_{\mathrm{nbr}}^{(\ell)} h_u^{(\ell)}$, $m_{v\to v}^{(\ell+1)} = W_{\mathrm{self}}^{(\ell)} h_v^{(\ell)}$. Thus, the source-node recursion is

$$r_{u\to v}^{(\ell+1)} = g_v^{(\ell+1)} \odot \left(W_{\mathrm{self}}^{(\ell)} r_{u\to v}^{(\ell)} + \frac{1}{|\mathcal{N}(v)|} W_{\mathrm{nbr}}^{(\ell)} \sum_{w\in\mathcal{N}(v)} r_{u\to w}^{(\ell)} \right).$$

II. Experimental Setup and Training Details

The experimental suite contains 17 tasks: six synthetic tasks, five real-world node-classification tasks, four molecular graph-classification tasks, and two regression tasks. B-XAIC indole, PAINS, and rings-count use three structural labeling rules on the same molecular collection. The synthetic suite includes BABiFanRole and BACycle alongside BAPathRole, BATriadic, BAHouse, and BAPath. This section specifies dataset construction, backbone training, SAE configurations, and validation selection.

A. Synthetic Dataset Construction

The synthetic suite contains three node-classification tasks and three graph-classification tasks. Each task supplies prediction labels and structural reference masks for recovery evaluation.

a) Node-classification tasks: We use NetworkX generators[1] with custom labels and motif attachments. BATriadic uses `powerlaw_cluster_graph` with $(n, m, p, \mathrm{seed}) = (2000, 3, 0.35, 37)$. Labels bucket the degree/triangle rank score with weights $0.65/0.35$. BAPathRole and BABiFanRole attach 250 copies of P_5 and $K_{2,3}$, respectively, to BA backbones with $(n, m) = (750, 3)$ and seeds 43 and 47. Their labels distinguish background and two motif roles. Anchors mix degree-plus-one and uniform sampling with weights $0.55/0.45$. Features combine structural statistics and Gaussian coordinates. All synthetic node datasets use stratified training, validation, and test splits of 60%, 20%, and 20%, respectively.

b) Graph-classification tasks: BA graph instances use $m = 2$ and n uniform on $\{22, \ldots, 38\}$, after which the generator inserts motifs. BAHouse, BAPath, and BACycle classify counts $0/1/2$ of house, P_5, and six-cycle motifs, respectively. Their seeds are 101, 404, and 202. Each class has 300 graphs, with six structural node features and stratified training, validation, and test splits of 70%, 15%, and 15%, respectively. Planted node and edge masks are stored separately for evaluation.

[1] NetworkX graph generators: https://networkx.org/documentation/stable/reference/generators.html.

B. *Backbone Training and Model Selection*

Backbone predictors are trained on the designated training splits using GCN, GIN, GAT, and GraphSAGE, with relation-aware or heterogeneous implementations where appropriate. A common validation-based procedure selects architecture, optimization settings, and checkpoints before explanation evaluation. Within each dataset–predictor case in Table IV of the main paper, SAE and the comparison methods share the same selected frozen predictor, data splits, and test targets.

The backbone search varies hidden width, message-passing depth, dropout, learning rate, and weight decay, with task-specific normalization and class weighting. Classification uses cross-entropy or binary cross-entropy; regression uses mean squared error. Validation macro-F1 selects B-XAIC configurations, validation average precision selects GraphLand node-classification, Web-Fraud, and IEEE-CIS configurations, and validation R^2 selects regression configurations. The reported searches use seed 42 for configuration selection and seed 0 for the final predictor fit. The selected predictor remains frozen during representation extraction, SAE training, and explanation.

C. *SAE Training and Selection Details*

The fixed-configuration SAE ablations use normalized Top-K codes with $K = 16$, an eightfold dictionary expansion, Adam with learning rate 10^{-3}, and 100 training epochs. The objective combines reconstruction, grounding, prediction faithfulness, and decoder coherence. In the full fixed setting, the respective weights are 1, 0.01, 0.1, and 0.01; magnitude and stability terms have zero weight. The three SAE seeds are 0, 1, and 2. The frozen predictor and train-only coordinate-wise normalizer are shared across the corresponding comparisons; decoder columns are normalized after each update. Grounding, faithfulness, and coherence weights follow the five-epoch ramp described in the main paper.

a) Regression grounding groups: For the two regression tasks, grounding uses deterministic structural groups. For city-roads-M, let $d_{\text{in}}(v)$ be the number of stored directed edges whose destination is node v, counted on the original input graph before model-specific propagation or explanation interventions. Its group is

$$g_v = \lfloor \log_2(1 + d_{\text{in}}(v)) \rfloor .$$

For RelSC (rdf), let $\mathcal{A}$ be the node-type set of a graph G and $V_a(G)$ its nodes of type a. We count all node types and assign

$$n_G = \sum_{a \in \mathcal{A}} |V_a(G)|, \qquad g_G = \lfloor \log_2(1 + n_G) \rfloor .$$

Bucket b contains structural counts from $2^b - 1$ through $2^{b+1} - 2$. The SAE training populations occupy three degree buckets (IDs $1, 2, 3$) for city-roads-M and nine graph-size buckets (IDs $4, \ldots, 12$) for RelSC. These analytical bin boundaries and group assignments are shared across the four backbone families. Grounding is computed from training representations and their structural group IDs. The continuous regression response is used for predictor training, and output-space MSE supplies the SAE prediction-faithfulness term.

b) Common parameter-selection protocol: The explanation results in Table IV of the main paper use a common validation-based parameter-selection protocol for SAE and the comparison methods. Method-specific training and extraction settings are selected on validation data, then fixed for test evaluation on the shared frozen predictor. Classification selection uses prediction faithfulness and available structural recovery; regression selection uses absolute prediction change. Explanation size is recorded with the resulting metrics. The same selected policy supplies all reported test metrics for each method–predictor case.

For SAE, the selected settings comprise auxiliary loss weights, dictionary expansion, active-feature count, learning rate, training duration, and source-score extraction. Training uses standardized training representations with a train-fitted normalizer. Extraction considers activity-weighted or prediction-weighted source scores, thresholds, and ranked prefixes. Validation selection precedes three-seed test evaluation. The fixed-budget suite follows the same validation-based selection principle and uses one selected ranking policy per predictor at all three budgets $8, 12, 16$. The fixed-configuration ablations retain the settings specified above.

III. Evaluation Protocol and Fixed-budget Results

A. *Native Deletion Intervention*

For selected source nodes $U \subseteq V$, the input-level deletion intervention is $\mathcal{I}(G, U) = (V, E', X')$, where

$$E' = \{(u, v) \in E : u \notin U,\ v \notin U\}, \qquad x'_v = \begin{cases} 0, & v \in U, \\ x_v, & v \notin U. \end{cases}$$

The intervention retains the vertex set, node indices, and graph-batch assignments, zeros the selected input vectors, and removes their incident edges. For large-graph adapters, zeroing is applied at the predictor's encoded input, before its fixed bias and message-passing layers. For heterogeneous graphs, U contains typed node IDs, and incident edges are removed from every relation while preserving all node and relation types.

The frozen predictor is reevaluated on this perturbed input. GCN degree normalization is recomputed from the perturbed graph, including global degree changes when a local query evaluator is used; GraphSAGE neighborhood means and GAT attention coefficients are likewise recomputed. Each backbone retains its native self-loop, root-message, bias, and readout rules. In particular, layers may reintroduce self-loops, and zero-input isolated nodes can acquire nonzero hidden states through biases or root transformations. Graph-level readout still includes the original node slots, with the original denominator for mean pooling. Model parameters and fitted feature-normalization statistics remain fixed.

For node predictions, U excludes the target node. Classification effects use the original predicted class for both original and perturbed outputs; regression effects compare

scalar predictions in the task's evaluation space. Explainers share this input intervention within each matched predictor comparison. Source decomposition uses the original forward-pass gates and attention coefficients, while deletion evaluation recomputes the perturbed prediction. Latent-feature ablation modifies SAE coefficients (main paper, Section IV-E).

B. Evaluation Metrics

Let E_τ denote the selected sources and G_τ the annotated reference set. Recovery is evaluated only where a reference is available:

$$\text{Precision}_\tau = \frac{|E_\tau \cap G_\tau|}{|E_\tau|}, \tag{1}$$

$$\text{Recall}_\tau = \frac{|E_\tau \cap G_\tau|}{|G_\tau|}, \tag{2}$$

$$\text{Jaccard}_\tau = \frac{|E_\tau \cap G_\tau|}{|E_\tau \cup G_\tau|}. \tag{3}$$

A zero denominator gives zero recovery. Recovery averages include tasks with available reference masks. The native-size table reports Precision/Jaccard, and the fixed-budget table reports Recall/Jaccard, with Rec. denoting recall. B-XAIC uses dataset-supplied structural annotations, and MUTAG188 uses chemical-rule proxy masks.

For B-XAIC, the GT masks in Table IV of the main paper are the task-specific annotations supplied with the dataset. The atom indices for indole, PAINS, and rings-count are read from the corresponding fields in `explanations.sdf`, preserving the original atom indexing. The benchmark defines both task labels and reference masks through structural rules.

For MUTAG188, atom-level proxy masks are constructed from chemical-substructure matching rules adapted from GraphXAI. In the hydrogen-suppressed graph, the NH_2 rule selects a nitrogen atom of degree one. The NO_2 rule selects a nitrogen atom of degree three together with its terminal oxygen neighbors when at least two such neighbors are present. Matching uses atom types and connectivity. These constructed masks supply the reference sets for MUTAG188 recovery metrics.

Dataset-supplied and proxy molecular masks use the same precision and Jaccard formulas. The Full mask contains all eligible source nodes and serves as the all-source deletion reference. GT, proxy, Full, and selected explanation masks are evaluated with the same deletion metrics. Real-world node tasks with prediction labels alone are evaluated through deletion faithfulness and explanation size.

The main paper's native-size results average over matched test targets and selected backbones, with SAE results averaged over seeds 0–2. Bold values mark the best displayed result in each dataset, including ties. Empty annotations or selections receive zero recovery. Section III-E specifies the validation-based selection procedures for the complete explanation pipelines.

Log./Pr. measure the decrease of the original target-class logit/probability after input deletion. Native output conventions are preserved, including binary $[0, z]$ logits, for which a zero class-0 logit decrease can accompany a nonzero probability decrease. Negative decreases indicate score increases after deletion. Regression reports the absolute prediction change $|f(G) - f(\mathcal{I}(G, E_\tau))|$ in the task's evaluation target space. Logit and regression changes are reported on each dataset's native scale.

C. Baseline Scoring Adaptations

Fixed-budget PGExplainer ranks nodes by incident-edge scores, and GraphSHAP-IQ uses signed first-order node scores. GECo-A broadcasts target-conditioned community readout scores to source nodes. GECo-L uses CNM for small fields and deterministic Louvain for large IEEE-CIS/Web-Fraud fields. For regression, community scores measure closeness to the original prediction, scaled by the training-prediction standard deviation. The fixed-budget suite uses the corresponding saved explainers and masks; native-size extraction is specified in Section III-D.

The qualitative examples in the main paper visualize selected source-node sets and their localization within the input graph.

D. Explainer Implementations and Node-set Conversion

Each comparison uses a frozen predictor and a common source-node deletion operation. Method-specific explanations are converted to eligible node sets before measuring their effects. The node being explained is excluded from the removable sources. Original node and relation types are retained in heterogeneous predictor evaluations. In Table IV of the main paper, PGExplainer and GECo denote the task-specific configurations and adaptations described below. The reference masks and recovery metrics are described in Section III-B.

GNNExplainer optimizes a target-specific soft mask. For node tasks, native-size conversion retains nodes with positive source scores. For graph tasks, it retains eligible endpoints of undirected edge pairs whose mean mask is at least 0.5, with a top-eight-edge-pair fallback. Explanation size is the number of distinct eligible endpoints. GraphSHAP-IQ uses signed first-order node scores: native-size extraction retains positive support, and fixed-budget extraction selects the highest-ranked eligible sources.

PGExplainer uses a shared edge-mask predictor. The native PGExplainer-N configuration uses mean edge-probability size regularization. Directed nonself edges meeting the selected probability threshold contribute their eligible endpoints, and the union of these endpoints defines the output node set. The standard threshold is 0.5; output size follows the thresholded edge support, with empty sets retained. The IEEE-CIS configuration PGExplainer-N-F uses a validation-selected threshold of 0.1 and size coefficient of 0.01.

GECo evaluates graph communities by their support for the frozen prediction and combines selected communities. GECo-A retains the target node during community evaluation. Classification scoring uses the original predicted-class probability, and regression scoring uses closeness to the original scalar prediction scaled by the training-prediction standard

deviation. Large IEEE-CIS and Web-Fraud fields use the GECo-L Louvain frontend with resolution one and seed 42. Community discovery uses an undirected projection where specified, and predictor evaluation retains the original typed relations.

SAE-based explanations use activity-weighted or prediction-weighted projections of source contributions onto the learned decoder directions. Native-size extraction considers positive-score support, relative thresholds, ranked prefixes, and the specified greedy refinements. The validation-selected complete policy supplies all test metrics. In the fixed-budget comparison, source ranking is converted to exactly $\min(k, n_\tau)$ eligible nodes using the common budget convention described in Section III-F.

E. SAE Extraction and Large-graph Optimization

The SAE extraction settings in Table IV follow the common validation-based protocol in Section II-C. The trained dictionary, coordinate-wise normalizer, and frozen GNN are reused across targets. For each target, extraction computes source scores, forms an initial node set, and applies the selected refinement policy. SAE results average seeds 0–2. The following steps describe our extraction procedure and its large-graph implementation.

a) Sparse candidate fields: For a node target, the source candidates are nodes in the predictor's message-passing receptive field, excluding the target itself. The evaluation context also includes nodes needed for degree normalization and typed messages. We store this field with sparse neighborhood structures and retain the mapping between local indices and original global node IDs. Graph-level explanations use the eligible nodes of the input graph. Static neighborhood indices, input features, and original-forward-pass predictor quantities are cached for source decomposition and repeated mask evaluation.

b) SAE source scoring: The target representation is encoded once, and source contributions are scored against its active decoder directions. For the activity-weighted score in Section III-D of the main paper, collect the decoder terms into a target-specific direction

$$a_\tau = \sum_{k \in \mathcal{K}_\tau} z_{\tau,k} \frac{d_k}{\|d_k\|_2}, \qquad S_\tau(u) = \langle c_{u;\tau} \oslash \rho, a_\tau \rangle .$$

This form separates decoder weighting from source scoring: the same a_τ is used for all candidate sources of target τ. Scores are mapped back to global node IDs before mask construction. The SAE sparsity parameter K specifies active dictionary features, and the extraction policy determines the number of selected input nodes.

c) Initial masks and local refinement: Initial masks use positive-score support, relative score thresholds, positive-score mass fractions, or ranked source prefixes, according to the validation-selected policy. City-reviews additionally considers complete one-hop neighborhoods and size-penalized masks under class-stratified validation; the one-hop candidates are defined by topology. Molecular and synthetic node explanations use the selected local mask refinement, while synthetic graph extraction uses relative positive-score thresholds. On RelSC, refinement makes up to 32 greedy additions or removals among SAE-scored sources and their one-hop neighbors, using absolute prediction change as the objective. The GNN and SAE remain frozen throughout extraction.

d) Block refinement on Web-Fraud: For the large Web-Fraud source field, candidates are grouped using SAE-score sign and quantiles, direct-neighbor membership, high-score source groups, and source-degree strata. Each proposed move adds or removes a whole block from the current mask. We evaluate the resulting mask by input deletion and accept moves that increase the decrease in the original predicted-class probability. Search performs at most eight improving moves and stops when the improvement is at most 10^{-6}. The final explanation contains the nodes in the selected blocks. One evaluated field contains 1,384,902 source candidates and 2,158,150 local edges.

e) Sparse deletion evaluation: Each candidate mask is evaluated on the sparse field using the intervention in Section III-A. Selected source inputs are zeroed and their incident edges removed, with node slots and relation types retained. The perturbed forward pass recomputes degree normalization, neighborhood means, attention weights, and activations according to the frozen backbone. For mean-GraphSAGE on Web-Fraud, surviving-neighbor features are accumulated in double precision and then passed through the original single-precision modules. Static indices and cached inputs are reused across candidate masks. The selected mask is mapped to the original graph and its deletion effect is checked by independent predictor replay.

F. Native-size and Fixed-budget Comparisons

The main paper's Table IV reports native-size pipelines, with output size recorded in the Nodes column. Table I and its continuation report fixed-budget rankings at $\min(k, n_\tau)$ eligible source nodes per target, where n_τ is the candidate count and $k \in \{8, 12, 16\}$. The node being explained is excluded from its source-node budget. Saved masks with fewer sources are padded to the required size under the fixed-budget protocol.

The fixed-budget suite uses four backbone families for each of 17 tasks, giving 68 dataset–backbone pairs. Native-size results use the selected predictor cohort and extraction settings for each dataset, including a single selected backbone for some large real-world tasks. Each comparison averages results within its own predictor cohort.

Fixed-budget explanation evaluation uses 1,600 target–predictor pairs. Cora uses 50 targets per backbone; synthetic node tasks and artnet-exp/city-reviews/city-roads-M use 25; synthetic graph tasks, molecular tasks, IEEE-CIS, Web-Fraud, and RelSC use 20. Predictor accuracy is evaluated on the full held-out population. Explanation metrics average SAE seeds 0–2 within targets, targets within predictors, backbones equally within datasets, and datasets equally within each synthetic task group.

TABLE I
FIXED-BUDGET EXPLANATION QUALITY.

(a) Synthetic

Node classification

Method	$k=8$				$k=12$				$k=16$			
	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑
Syn-Node												
GNNEx	1.994	**0.254**	0.440	0.204	1.994	0.251	0.491	0.231	1.968	**0.257**	0.538	0.258
GECo-A	1.074	0.108	**0.442**	0.207	1.255	0.128	0.489	0.236	1.423	0.147	0.532	0.259
PGExplainer	1.413	0.182	0.431	0.204	1.491	0.198	0.487	0.233	1.591	0.213	0.529	0.259
GraphSHAP-IQ	1.845	0.221	0.441	**0.210**	1.992	0.233	**0.494**	**0.238**	2.001	0.233	**0.548**	0.264
SAE (Ours)	**2.052**	0.250	0.428	0.205	**2.107**	**0.255**	0.489	**0.238**	**2.124**	0.255	0.538	**0.266**

Graph classification

Method	$k=8$				$k=12$				$k=16$			
	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑
Syn-Graph												
GNNEx	3.195	0.453	0.060	0.033	5.169	0.511	0.152	0.072	6.439	0.534	0.246	0.103
GECo	7.832	0.601	0.294	0.282	**11.329**	0.648	0.411	0.350	**12.341**	0.623	0.427	0.278
PGExplainer	2.730	0.456	0.143	0.089	3.778	0.499	0.184	0.094	4.408	0.503	0.212	0.088
GraphSHAP-IQ	7.942	0.600	0.178	0.154	10.017	0.620	0.256	0.183	11.286	0.623	0.335	0.183
SAE (Ours)	**9.163**	**0.698**	**0.475**	**0.422**	10.948	**0.678**	**0.585**	**0.435**	10.598	**0.633**	**0.592**	**0.331**

(b) Real-world

Node classification

Method	$k=8$		$k=12$		$k=16$	
	Log.↑	Pr.↑	Log.↑	Pr.↑	Log.↑	Pr.↑
Cora						
GNNEx	5.855	0.304	6.064	0.316	6.314	0.346
GECo-A	3.032	0.185	3.575	0.219	3.838	0.228
PGExplainer	1.908	0.103	2.574	0.123	2.931	0.143
GraphSHAP-IQ	6.477	**0.362**	6.812	0.388	7.066	**0.406**
SAE (Ours)	**6.486**	0.354	**6.932**	**0.395**	**7.076**	**0.406**
artnet-exp						
GNNEx	0.054	-0.017	0.061	-0.017	0.065	-0.019
GECo-A	-0.004	0.006	-0.002	0.007	-0.003	0.006
PGExplainer	-0.003	-0.005	0.002	-0.004	0.001	-0.004
GraphSHAP-IQ	0.053	0.012	**0.102**	0.019	0.105	0.018
SAE (Ours)	**0.056**	**0.023**	0.100	**0.030**	**0.107**	**0.034**
city-reviews						
GNNEx	0.014	0.003	0.033	0.005	0.036	0.004
GECo-A	-0.003	0.004	0.001	0.004	-0.000	0.004
PGExplainer	-0.002	-0.001	-0.002	-0.001	-0.006	0.002
GraphSHAP-IQ	0.018	-0.002	0.027	-0.000	0.032	-0.001
SAE (Ours)	**0.066**	**0.033**	**0.083**	**0.036**	**0.099**	**0.038**
IEEE-CIS						
GNNEx	0.004	0.012	0.079	0.034	0.107	0.041
GECo-L	0.229	0.060	0.240	0.060	0.255	0.067
PGExplainer	-0.095	-0.017	-0.157	-0.025	-0.150	-0.023
GraphSHAP-IQ	0.531	0.123	0.534	0.125	0.541	0.130
SAE (Ours)	**0.555**	**0.156**	**0.593**	**0.167**	**0.612**	**0.172**
Web-Fraud						
GNNEx	0.161	0.052	0.163	0.053	0.163	0.053
GECo-L	0.098	0.029	0.097	0.030	0.098	0.031
PGExplainer	0.104	0.035	0.100	0.033	0.097	0.035
GraphSHAP-IQ	**0.230**	**0.054**	**0.233**	**0.055**	**0.236**	**0.057**
SAE (Ours)	0.148	0.049	0.152	0.050	0.154	0.052

Graph classification

Method	$k=8$				$k=12$				$k=16$			
	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑	Log.↑	Pr.↑	Rec.↑	Jac.↑
B-XAIC indole												
GNNEx	**6.643**	0.795	0.368	0.269	**7.454**	0.812	0.560	0.350	**8.133**	0.815	0.721	0.389
GECo	5.098	0.647	0.370	0.317	6.612	0.764	0.499	0.343	7.706	0.817	0.655	0.378
PGExplainer	3.847	0.508	0.254	0.184	5.467	0.621	0.353	0.208	6.376	0.689	0.453	0.222
GraphSHAP-IQ	6.560	0.786	**0.489**	**0.408**	7.376	0.811	**0.653**	**0.446**	7.953	0.819	**0.737**	**0.407**
SAE (Ours)	6.448	**0.808**	0.448	0.356	7.186	**0.830**	0.572	0.371	7.797	**0.826**	0.670	0.361
B-XAIC PAINS												
GNNEx	2.452	0.557	0.342	0.243	3.302	0.595	0.511	0.299	3.688	0.609	0.657	0.332
GECo	2.946	0.622	**0.432**	**0.281**	3.427	**0.638**	0.562	0.310	3.724	**0.629**	0.694	0.340
PGExplainer	2.025	0.472	0.292	0.219	2.565	0.538	0.423	0.263	3.030	0.564	0.571	0.302
GraphSHAP-IQ	**3.148**	**0.631**	0.429	0.274	**3.571**	0.630	**0.587**	**0.321**	**3.860**	0.621	**0.715**	**0.347**
SAE (Ours)	2.584	0.551	0.392	0.277	3.101	0.573	0.536	0.311	3.356	0.579	0.640	0.318
B-XAIC rings-count												
GNNEx	**10.551**	0.718	0.216	0.211	**14.086**	0.758	0.324	0.310	**16.379**	0.783	0.424	0.396
GECo	5.909	0.656	**0.231**	**0.226**	8.297	0.729	**0.342**	**0.328**	10.580	0.743	**0.446**	**0.418**
PGExplainer	6.198	0.617	0.152	0.144	9.142	0.711	0.242	0.222	11.296	0.741	0.337	0.300
GraphSHAP-IQ	9.140	0.686	0.205	0.198	12.054	0.736	0.311	0.294	14.073	0.755	0.410	0.378
SAE (Ours)	8.477	**0.737**	0.195	0.187	11.412	**0.790**	0.292	0.272	13.769	**0.810**	0.381	0.344
MUTAG188												
GNNEx	**1.795**	0.287	0.212	0.090	1.848	0.340	0.333	0.119	1.746	0.373	0.628	0.178
GECo	1.171	0.320	0.312	0.177	1.438	0.370	0.413	0.185	1.681	0.385	0.698	0.223
PGExplainer	1.083	0.241	**0.628**	0.291	1.542	0.308	**0.761**	**0.265**	1.608	0.354	**0.867**	0.245
GraphSHAP-IQ	1.601	0.334	0.596	**0.312**	**1.872**	0.366	0.685	0.264	**1.824**	0.380	0.839	**0.254**
SAE (Ours)	1.546	**0.352**	0.244	0.163	1.586	**0.390**	0.323	0.165	1.672	**0.394**	0.618	0.204

Log./Pr.: native target-class logit/probability decrease; Rec./Jac.: recall/Jaccard. Every method selects exactly $\min(k, n)$ eligible sources, excluding the node target; n is the candidate count. Five unique explainers share all four frozen backbones per task and the original fixed-budget test targets (Cora: 50 per backbone). Means average SAE seeds 0–2, targets, backbones, then datasets equally. Syn-Node: BAPathRole, BATriadic, BABiFanRole; Syn-Graph: BAPath, BAHouse, BACycle; MUTAG recovery uses the chemical-rule proxy; recovery columns are reported for annotated tasks. SAE (Ours) uses validation-selected Top-K training and extraction settings, three seeds, and four active losses. One selected ranking policy per predictor supplies all budgets and metrics, following Section II-C. PGExplainer uses incident-edge ranking and GraphSHAP-IQ signed first-order ranking. GECo-A ranks target-conditioned community readout scores; GECo-L uses CNM on small fields and deterministic Louvain on large IEEE-CIS/Web-Fraud fields. Implementation and selection details are given in Sections III-D and II-C. Cyan: baselines; magenta: SAE (Ours); bold: displayed best values including ties.

Table II summarizes first-place counts over 36 task-group/dataset–metric combinations per budget: two synthetic groups with four metrics each, five real-world node datasets with two metrics each, four molecular tasks with four metrics each, and two regression tasks with one metric each. Counts use the displayed means, include all tied methods, and group GECo variants.

SAE attains 17 first-place values at each of $k = 8, 12, 16$. The corresponding counts are $4/3/3$ for GNNExplainer, $5/4/5$ for GECo variants, $1/2/1$ for PGExplainer, and $9/11/11$ for GraphSHAP-IQ. SAE has the largest first-place count at all three node budgets.

G. Fixed-budget Result Details

Table I reports classification faithfulness and structural recovery at three fixed node budgets. On the synthetic graph tasks, SAE achieves probability decreases of 0.698, 0.678, and 0.633 and Jaccard scores of 0.422, 0.435, and 0.331 at $k = 8, 12, 16$, respectively, leading both metrics at each budget. On city-reviews and IEEE-CIS, SAE has the largest logit and probability decreases at all three budgets. For B-XAIC indole, rings-count, and MUTAG188, it has the largest probability decrease at every budget. These results show effective prediction attribution under fixed output sizes.

Table I reports absolute prediction changes for the two regression tasks. On RelSC, SAE obtains 0.058, 0.075, and 0.089 at $k = 8, 12, 16$, compared with 0.045, 0.057, and 0.067 for GECo-A, the next-largest values. On city-roads-M, SAE obtains 0.303, 0.290, and 0.310, with the largest value at $k = 8$. The RelSC results place SAE first at all three budgets.

IV. COMPUTATIONAL COST

Table III reports measured fitting cost F, measured online explanation cost t, and modeled per-target cost $C_{100} = F/100 + t$. Online timing uses one predetermined target for each of 17 tasks and four backbones, giving 68 cases.

TABLE I
FIXED-BUDGET EXPLANATION QUALITY (CONTINUED).

(c) Regression

Method	Absolute prediction change ↑		
	$k = 8$	$k = 12$	$k = 16$
city-roads-M			
GNNEx	0.301	**0.315**	0.315
GECo-A	0.279	0.310	**0.325**
PGExplainer	0.223	0.300	0.295
GraphSHAP-IQ	0.166	0.261	0.284
SAE (Ours)	**0.303**	0.290	0.310
RelSC			
GNNEx	0.015	0.018	0.022
GECo-A	0.045	0.057	0.067
PGExplainer	0.007	0.008	0.008
GraphSHAP-IQ	0.031	0.038	0.045
SAE (Ours)	**0.058**	**0.075**	**0.089**

Methods share frozen predictors and test targets and select $\min(k, n)$ eligible sources. SAE (Ours) uses the validation-selected, three-seed protocol of the classification panel. GECo-A ranks community closeness $\exp(-|\hat{y}_C - \hat{y}|/\sigma_{\text{train}})$, where σ_{train} is the training-prediction standard deviation. PGExplainer uses incident-edge ranking and GraphSHAP-IQ signed first-order ranking. Bold marks the largest displayed values, including ties.

TABLE II
FIRST-PLACE COUNTS UNDER FIXED NODE BUDGETS.

Method	$k = 8$	$k = 12$	$k = 16$
GNNEx	4	3	3
GECo variants	5	4	5
PGExplainer	1	2	1
GraphSHAP-IQ	9	11	11
SAE (Ours)	**17**	**17**	**17**

Counts cover 36 displayed task-group/dataset–metric combinations per budget. Synthetic node and graph datasets are each aggregated as one group. Ties count for every tied method, and GECo variants are grouped. Full results and evaluation protocols are provided in Table I and Section III-F.

Measurements use cached inputs and one CPU thread on one physical core of an AMD EPYC 7K62 processor. The timed *SAE-less* uses the fixed 100-epoch configuration in Section II-C; the native-size quality results use the extraction policies in Section III-E.

TABLE III
MEASURED FITTING AND ONLINE EXPLANATION COSTS, WITH MODELED AMORTIZATION AT $N = 100$ (SECONDS). MEANS COVER 17 TASKS × FOUR BACKBONES (68 CASES), WITH ONE PREDETERMINED TARGET PER CASE.

Method	$\bar{F}\downarrow$	$\bar{t}\downarrow$	$\bar{C}_{100}\downarrow$
SAE-less	9.14	**2.097**	**2.19**
GNNExplainer	**0**	224.028	224.03
GECo	**0**	2.781	2.78
PGExplainer	—	2.710	—
GraphSHAP-IQ	**0**	192.004	192.00

SAE fitting averages three seed fits per case. Zero F denotes methods with target-specific computation and no shared explainer training. PGExplainer has complete fitting measurements for 9 of 68 cases; dashes denote unavailable full-cohort measured fitting costs. C_{100} is calculated from the measured components. Bold marks column minima, including ties.

SAE has a mean online time of 2.097 seconds, compared with 224.028 for GNNExplainer, 2.781 for GECo, 2.710 for PGExplainer, and 192.004 for GraphSHAP-IQ. Its mean fitting cost is 9.14 seconds. Applying $F/N + t$ gives per-target costs of approximately 11.24, 3.01, 2.19, and 2.11 seconds at $N =$ 1, 10, 100, and 1,000, respectively. SAE has the lowest mean online time in the 68-case timing suite, and its modeled fitting contribution decreases with dictionary reuse.

A. Fitting and Online Cost Definitions

For method m on dataset–backbone case d, let $F_{d,m}$ be the one-time fitting cost and $t_{d,m,i}$ the online cost of explaining target i. The per-explanation cost over N targets is

$$C_{d,m}(N) = \frac{F_{d,m} + \sum_{i=1}^{N} t_{d,m,i}}{N}. \tag{4}$$

The reuse curves substitute each case's measured single-target time $t_{d,m}$ into $C_{d,m}(N) = F_{d,m}/N + t_{d,m}$. SAE and PGExplainer have a shared fitting stage. SAE fitting includes initialization and 100 training epochs on cached training representations; the three separately timed seed fits are averaged within each case to obtain the cost of one SAE fit.

Online timing starts before target-specific score generation and stops after top-16 node selection. For SAE, this includes sparse coding, source-contribution decomposition, and decoder projection. GNNExplainer mask optimization, GECo community scoring, and GraphSHAP-IQ coalition evaluation are included in their respective online costs. Predictor training, configuration search, loading, shared representation extraction, static index/candidate preparation, induced-edge materialization, and deletion-based evaluation are outside the timed stages.

GraphSHAP-IQ timing uses first-order scores from truncated coalition calculations with efficiency correction and optimized recursive singleton evaluation on applicable backbones. Ten of the 68 cases use the constant-zero-game shortcut. Across the 68 cases, mean online time is 192.004 seconds, the median is 0.232 seconds, and the maximum is 6,250.376 seconds. Separately recorded engine setup has a mean of 0.705 seconds and a maximum of 25.349 seconds; it precedes the online timer.

B. Timing Measurements and Fitting Estimates

All five methods have online measurements for the 68 task–backbone cases. SAE fitting measurements cover 204 seed fits under the same CPU setting. Three seed fits are averaged within each case, followed by equal averaging over cases.

PGExplainer fitting records comprise nine complete runs, 16 partial runs, and 43 short probes. Table III reports its measured online mean and marks the full-cohort measured fitting cost with a dash. The main paper's runtime curve uses the fitting estimate defined below.

The PGExplainer fitting estimate uses the nine complete fitting times, scales measured optimizer-step durations from the 16 partial runs to their configured workloads, and adds initialization to scaled short-step times for the 43 probes.

TABLE IV
MEASURED SAE FITTING AND ONLINE EXPLANATION COMPONENTS BY BACKBONE (SECONDS).

Backbone	SAE $\bar{F}$	SAE $\bar{t}$	PG $\bar{t}$
GCN	6.24	1.697	3.501
GIN	8.98	3.242	2.638
GAT	14.62	3.027	4.116
GraphSAGE	6.73	0.422	0.583

Arithmetic means over 17 tasks per backbone, rounded for display. SAE fitting averages three seeds within each task. Online columns use one predetermined target per case.

Averaging over the 68 cases gives $\bar{\bar{F}}_{\mathrm{PG}} = 19{,}034.536$ seconds. The modeled curve is $\bar{\bar{F}}_{\mathrm{PG}}/N + \bar{t}_{\mathrm{PG}}$, where the measured online mean is $\bar{t}_{\mathrm{PG}} = 2.709537$ seconds. The analytical reuse axis extends to $N = 10^6$.

C. Backbone Results and Aggregation

Table IV reports fitting and online costs by backbone. SAE fitting ranges from 6.24 seconds for GCN to 14.62 seconds for GAT. Its online means are 1.697, 3.242, 3.027, and 0.422 seconds for GCN, GIN, GAT, and GraphSAGE, respectively. SAE has lower mean online cost than PGExplainer on GCN, GAT, and GraphSAGE.

Aggregate components weight the 68 dataset–backbone cases equally, giving $\bar{C}_m(100) = \bar{F}_m/100 + \bar{t}_m$. For SAE, $\bar{F} = 9.142673$ and $\bar{t} = 2.097057$ seconds yield $\bar{C}_{100} = 2.188484$ seconds.

V. CONTROLLED ABLATIONS AND REPRESENTATION METRICS

The loss and sensitivity experiments use the fixed 100-epoch SAE configuration across 17 tasks, four backbone families, and three seeds. Each setting uses the same populations of at most 2,048 training representations and at most 2,048 representations per validation/test split. The coordinate-wise normalizer is fitted on training representations. Each auxiliary-loss ablation sets one coefficient to zero; the reconstruction-only variant sets all three auxiliary coefficients to zero. Fixed-budget pipeline selection follows the separate procedure in Section II-C.

Sensitivity plots report Purity@5 and prediction agreement for classification tasks. Purity@5 is the majority ground-truth class fraction among the five largest activations of each eligible feature, averaged over eligible features. Eligibility requires at least three training activations and five activations in the evaluated population, with activity threshold 10^{-8}. Undefined purity values are omitted from the average. Agreement is the fraction of reconstructed-representation predictions matching the original frozen predictor. The main paper's feature-alignment results use the node- and graph-specific purity definitions given in its Section IV-E.

Reconstruction MSE is computed in standardized representation coordinates. Reconstruction NMSE divides MSE by the mean coordinate variance in the evaluated population. Classification prediction fidelity uses KL divergence between original and reconstructed prediction distributions. Regression fidelity uses output MSE, normalized by the variance of the original predictions where specified. Structural recovery uses the node-set metrics in Section III-B.

A. ℓ_1 SAE+Path *Baseline*

The ℓ_1 *SAE+Path* baseline trains a ReLU SAE on standardized representations from a frozen GNN, using an activation sparsity penalty. For a minibatch $\mathcal{B}$, its training objective is

$$\mathcal{L}_{\mathrm{rec}} + \lambda_1 \frac{1}{|\mathcal{B}|d} \sum_{\tau \in \mathcal{B}} \sum_{j} z_{\tau,j} \|d_j\|_2,$$

where $\mathcal{L}_{\mathrm{rec}}$ is the mean squared reconstruction error, d is the representation dimension, $z_{\tau,j} \geq 0$ is the activation of feature j, and d_j is its decoder column. Decoder columns are normalized after each optimization step. The objective contains reconstruction and the ℓ_1 sparsity term.

The dictionary has eight times the representation width. Training uses Adam with learning rate 0.001, minibatches of 512, and 100 epochs. For each dataset–backbone case, the coefficient is selected from $\{0.001, 0.01, 0.1, 1\}$ using seed 0 validation results before test evaluation. Candidates with mean active-feature count between 12 and 20 are preferred, with the lowest validation NMSE selected among them. If none meets this range, the candidate closest to 16 in absolute log-ratio is selected. The selected coefficient is reused for the other two seeds.

For explanation, source-node contributions are obtained using the decomposition in Section III-C of the main paper and projected using the trained SAE features and decoder directions. The baseline and Top-K comparator use the same path-based mapping. The comparison covers six tasks, four backbones, and three SAE seeds, with explanation budget $k = 12$. The task-specific evaluation populations and native values are described in the following subsection.

B. Sparse-encoder Comparison Results

Table V compares the full Top-K SAE, a reconstruction-only Top-K SAE, and the ℓ_1 SAE using the same path mapping and explanation budget $k = 12$. The comparison covers BAPathRole, BATriadic, Cora, BAHouse, B-XAIC indole, and MUTAG188, with four backbones and three SAE seeds. Node and graph purity each average three datasets equally after seed/backbone averaging. Probability decrease averages all six tasks, and Jaccard averages the five annotated tasks.

The full Top-K SAE achieves node and graph purity of 87.84% and 69.71%, compared with 66.66% and 64.30% for the ℓ_1 SAE. Their measured active-feature counts are 16.0 and 104.7, respectively. Full Top-K obtains a probability decrease of 44.62%, Jaccard of 26.27%, reconstruction NMSE of 0.0517, and prediction KL of 0.0138. The reconstruction-only Top-K variant has prediction KL of 0.0159. Full Top-K therefore provides a more compact active representation than the ℓ_1 SAE and the lowest prediction KL among the three variants.

TABLE V
SPARSE-ENCODER RESULTS WITH A COMMON PATH MAPPING AND NODE BUDGET $k = 12$.

Metric	Full Top-K	Rec.-only	ℓ_1 SAE
Node purity (%) ↑	87.84	**87.85**	66.66
Graph purity (%) ↑	**69.71**	69.65	64.30
Pr. decrease (%) ↑	**44.62**	44.54	44.46
Jaccard (%) ↑	26.27	**26.44**	26.21
Active features ↓	**16.0**	**16.0**	104.7
Reconstruction NMSE ↓	0.0517	**0.0506**	0.0566
Prediction KL ↓	**0.0138**	0.0159	0.0265

Rec.-only uses a reconstruction-only Top-K SAE. Full Top-K uses reconstruction and the three auxiliary objectives. Bold marks the best displayed values, including ties. Purity uses the main paper's node- and graph-specific definitions. Active-feature counts are measured after training.

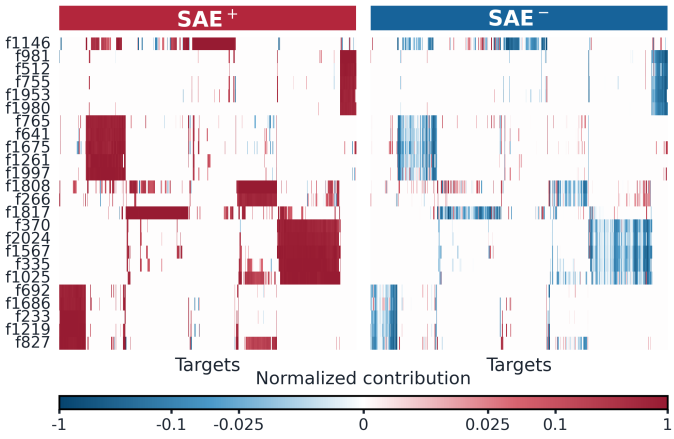


Fig. 1. Positive and negative source contributions to SAE features for GraphSAGE on Cora. Rows are SAE features, and columns are test targets. Red and blue indicate positive and negative normalized contributions, respectively.

C. *Positive and Negative Evidence Comparison*

The matched-budget comparison uses fixed 100-epoch Top-K ($K = 16$, eightfold dictionary) checkpoints, three SAE seeds, four frozen backbones, and the original 1,420 test targets across 15 classification tasks. Means average seeds, targets, backbones, and then datasets. Recovery averages the ten annotated tasks. Every variant selects $\min(B, n_\tau)$ eligible sources. Mixed evidence allocates half the budget to each sign and fills shortages by remaining absolute score. Single-sign rankings use deterministic zero-score padding.

Figure 1 displays feature-level contributions from positive- and negative-score sources for 630 correctly predicted Cora targets under GraphSAGE. Each target has at least three sources of each sign. The 24 displayed features have the largest mean absolute class-level contributions across the two source groups. For each target, *SAE+* sums contributions from the three highest-scoring positive sources, and *SAE-* uses the three lowest-scoring negative sources. Both panels share the feature and target order. Contributions are normalized by the largest absolute grouped contribution for each target across the two panels and displayed on a symmetric logarithmic scale. *SAE+* forms localized positive blocks, and *SAE-* is predominantly negative. Source scores sum contributions over active features, with individual feature contributions retaining their own signs.

TABLE VI
POSITIVE, NEGATIVE, AND BALANCED MIXED SAE EVIDENCE AT MATCHED NODE BUDGETS.

B	Variant	Log.↑	Pr.(%)↑	Prec.(%)↑	Jac.(%)↑
8	*SAE+*	**3.92**	**37.14**	**47.37**	**28.11**
	SAE-	1.90	24.65	29.33	14.24
	SAE+/-	3.13	32.86	41.14	21.95
12	*SAE+*	**4.55**	**38.16**	**43.93**	**31.12**
	SAE-	2.40	27.81	28.91	16.55
	SAE+/-	3.93	35.86	40.02	25.48
16	*SAE+*	**4.71**	**37.72**	**40.18**	**29.92**
	SAE-	2.75	29.90	29.09	18.83
	SAE+/-	4.55	36.83	39.07	28.00

Means cover 15 classification tasks and four backbones. Recovery uses the ten annotated tasks. Bold marks the largest mean within each budget.

Table VI compares positive, negative, and balanced mixed evidence at matched node budgets. At $B = 12$, *SAE+* obtains a logit decrease of 4.55, probability decrease of 38.16%, precision of 43.93%, and Jaccard of 31.12%. Its probability decrease and Jaccard exceed mixed evidence by 2.30 and 5.64 percentage points, respectively. *SAE+* has the largest mean in all four metrics at each of $B = 8, 12, 16$, supporting its use as the default evidence selection.

VI. SENSITIVITY ANALYSIS

This section reports single-factor sensitivity on the test split and joint grounding–faithfulness sensitivity on the validation split. The experiments vary auxiliary loss weights and dictionary width under the fixed SAE training protocol in Section V.

A. *Individual Weights and Dictionary Width*

Figure 2 reports test-set sensitivity to auxiliary weights (0.1, 0.3, 1, 3, and 10 times defaults) and dictionary width (2–16 times input dimension). Means average three seeds and four backbones within each of 15 equally weighted classification datasets. Bands are 95% Student-t intervals across dataset means.

Agreement remains near 97–98%. The grounding sweep raises Purity@5 from approximately 87.5% to 89.9%; faithfulness reaches 88.4%, and coherence remains near 88.0%. Sixteenfold width gives purity of 87.9% and agreement of 97.6%. These settings maintain high agreement while grounding adjusts feature purity.

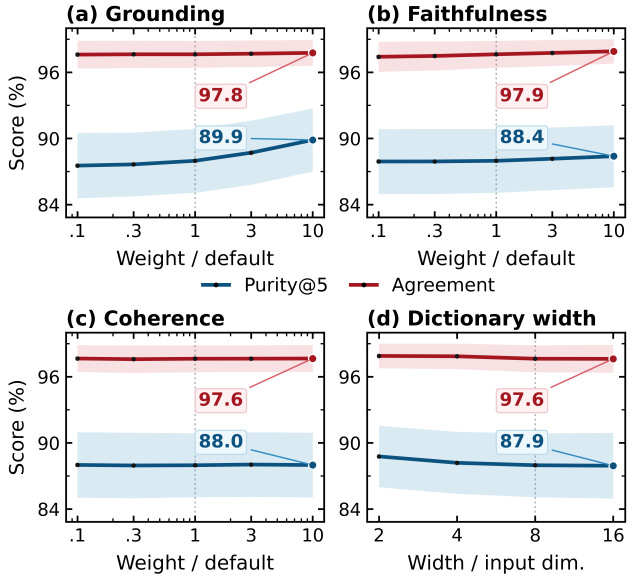


Fig. 2. Test-set sensitivity to auxiliary weights and dictionary width. Blue: Purity@5; red: agreement. Dashed lines mark defaults. Bands are 95% Student-t intervals across 15 dataset means.

B. *Joint Grounding and Faithfulness Weights*

Figure 3 reports joint validation sensitivity at 0.1, 1, and 10 times the default grounding and faithfulness weights. Tenfold grounding gives Purity@5 of 90.4–90.5%, compared with 87.8–88.7% at lower settings. Tenfold faithfulness gives agreement of 97.9–98.0% and prediction KL of 9.63–9.93×10^{-3}. Across the grid, reconstruction MSE is 0.075–0.079, probability drop is 26.9–27.2 percentage points, and Jaccard is 19.1–19.4%. Downstream explanation metrics remain stable across these settings.

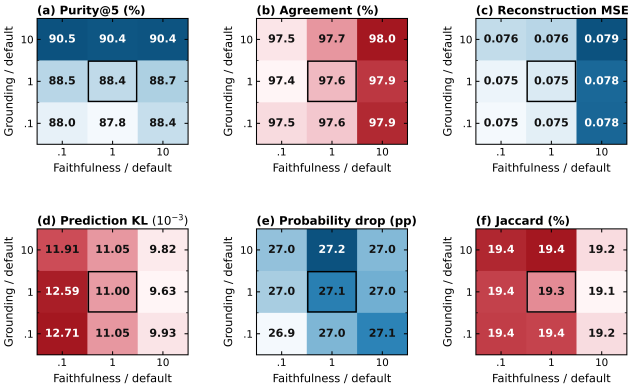


Fig. 3. Joint validation sensitivity to grounding and faithfulness weights. Multipliers are relative to defaults; black boxes mark the default pair.